<table>
<tr><th>Section</th><th>Ask</th><th>Points</th><th>What good looks like</th><th>What average looks like</th><th>What poor looks like</th><th>What nothing looks like</th><th>% Weightage</th></tr>
<tr><td></td><td></td><td>60</td><td>80-100%</td><td>60-80%</td><td><60%</td><td>0%</td><td></td></tr>
<tr><td>Exploratory Data Analysis</td><td>- Problem definition, questions to be answered
- Data background and contents
- Univariate analysis
- Bivariate analysis
- Key meaningful observations on individual variables and the relationship between variables</td><td>8</td><td>1) Definition of the problem (as per given problem statement with additional views) [0.5]

2) Observations on the shape of data, data types of various attributes, missing values, statistical summary. [1]

3) Univariate Analysis (Boxplots and distribution plots for numerical data and bar plots for categorical data) [1]

4) Bivariate Analysis (Correlation matrix, bivariate analysis for categorical and numerical variables with the target variable) [3]

5) Comments on the range of attributes, outliers of various attributes [1]

6) Insights from all the graphs plotted [1.5]</td><td>1) Definition of the problem (as per given problem statement with additional views)

2) Observations on the shape of data, data types of various attributes, missing values, statistical summary.

3) Univariate Analysis (Boxplots and distribution plots for numerical data and bar plots for categorical data)

4) Bivariate Analysis (Correlation matrix)

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2) Univariate Analysis (Boxplots and distribution plots for numerical data and bar plots for categorical data)

3) Bivariate Analysis (Correlation matrix)</td><td>No code/report</td><td>12.50%</td></tr>
<tr><td>Data Preprocessing</td><td>- Prepare the data for analysis
- Feature Engineering
- Missing value Treatment
- Outlier Treatment

- Ensure no data leakage among train-test and validation sets</td><td>5</td><td>1) Detect and treat missing values in Education_Level, Marital_Status and Income_Category and treat them [2]

2) Missing value treatment done after train, test and validation split [1]

3) Encoded Attrited customer as 1 and Existing customer as 0 in the target variable [1]

4) Test set should only be used at the end after getting a final model [1]

Note: Learners can do either of the following:
a) Either Split dataset into train/test sets. Train model with cross validation on trainset and report cross validation result for model selection
b) Split dataset into train/val/test sets. Train model on trainset and report validation result on validation set for model selection</td><td>1) Detect missing values in Education_Level, Marital_Status and treat them

2) Missing value treatment should be done after train, test and validation split

3) Oversampling and undersampling should be done on only train set

4) Encoded Attrited customer as 1 and Existing customer as 0 in the target variable</td><td>1) Detect missing values in Education_Level, Marital_Status and treat them

2) Missing value treatment should be done after train, test and validation split</td><td>No code/report</td><td>7.50%</td></tr>
<tr><td>Model Building - Original Data</td><td>- Choose the appropriate metric for model evaluation
- Build 5 models (from decision trees, bagging and boosting methods)
- Comment on the model performance

* You can choose NOT to build XGBoost if you are facing issues with the installation</td><td>6</td><td>1) Choose the appropriate metric for model evaluation [1]

2) Build 5 classification models, train them using the original train data, and comment on the model performance [1x5]</td><td>1) Build 2-4 classification models and trained them using the original train data</td><td>1) Build < 2 classification models and trained them using the original train data</td><td>No code/report</td><td>7.50%</td></tr>
<tr><td>Model Building - Oversampled Data</td><td>- Oversample the train data
- Build 5 models (from decision trees, bagging and boosting methods)
- Comment on the model performance

* You can choose NOT to build XGBoost if you are facing issues with the installation</td><td>6</td><td>1) Oversample the train data [1]

2) Train 5 classification models using the oversampled train data and comment on the model performance [1x5]</td><td>1) Oversample the train data

2) Fit 2-4 tuned models on oversampled data</td><td>1) Oversample the train data

2) Fit < 2 tuned models on oversampled data</td><td>No code/report</td><td>10.00%</td></tr>
<tr><td>Model Building - Undersampled Data</td><td>- Undersample the train data
- Build 5 models (from decision trees, bagging and boosting methods)
- Comment on the model performance

* You can choose NOT to build XGBoost if you are facing issues with the installation</td><td>6</td><td>1) Undersample the train data [1]

2) Train 5 classification models using the undersampled train data and comment on the model performance [1x5]</td><td>1) Undersample the train data

2) Fit 2-4 tuned models on undersampled data</td><td>1) Undersample the train data

2) Fit < 2 tuned models on undersampled data</td><td>No code/report</td><td>10.00%</td></tr>
</table>

Model Performance Improvement using Hyperparameter Tuning	- Choose 3 models (at least) that might perform better after tuning with proper reasoning - Tune the 3 models (at least) obtained above using randomized search and metric of interest - Comment on the performance of 3 tuned models * You can choose NOT to tune XGBoost if you experience long runtimes	**13**	1) Choose at least 3 best models from the ones built previosuly with proper reasoning [1] 2) Tune the best 3 models obtained above using randomized search and metric of interest [3x3] 3) Check and comment on the performance of 3 tuned models [1x3]	1) Explained model evaluation criteria and selected any other metric as metric of interest 2) Choose 3 best models 3) Tune the best 3 models obtained above using randomized search and metric of interest	1) Choose 3 best models 2) Tune the best 3 models obtained above using randomized search and metric of interest	No code/report	17.50%
Model Performance Comparison and Final Model Selection	- Compare the performance of tuned models - Choose the best model - Comment on the performance of the best model on the test set	**4**	1) Compare the performance of tuned models [2] 2) Choose the best model from the tuned models with proper reasoning [1] 3) Check and comment on the performance of the best model on the test set [1]	1) Compare model performances of tuned models 2) Choose the best model 3) Find test performance	1) Compare model performances of tuned models 2) Choose the best model	No code/report	10.00%
Actionable Insights & Recommendations	- Write down insights from the analysis conducted - Provide actionable business recommendations	**4**	1) Insights (at least 4) [0.5x4] 2) Recommendations (at least 2) [1x2]	1) Recommendations provided but are not actionable.	1) Recommendations - irrelevant to the business problem	No code/report	7.50%
Presentation / Notebook - Overall Quality	- Structure and flow - Crispness - Visual appeal - Conclusion and Business Recommendations OR - Structure and flow - Well commented code - Conclusion and Business Recommendations	**8**	1) Clear structure and flow - everything sits well in a story 2) Crispness (Not too many words, just enough to keep the focus on key things/points) 3) Visual appeal (Use of charts, colors, diagrams, format, symmetry, informative visualizations that are easy to interpret) OR 1) Well structured notebook with a logical flow 2) Clean and well commented code	1) There is structure and flow but some bits are jumbled 2) Points are made but in too many words 3) Lesser charts, format is not the cleanest OR 1) There is structure and flow but some bits are missing 2) Some of the code is commented * If any section is missing and points have been deducted for the same in a previous section, then no points should be deducted in this section	1) No structure or flow 2) Only a few points are covered, story is not complete 3) Not many visuals used OR 1) no structure or flow 2) no comments in the code	No report OR No code	10.00%